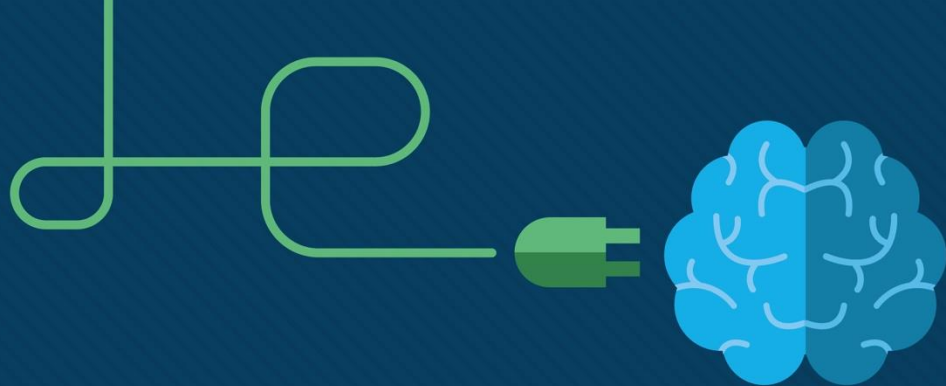




# Chapter 8: Subnetting IP Networks

CCNA Routing and Switching  
Introduction to Networks v6.0



# Chapter 8 - Sections & Objectives

- 8.1 Subnetting an IPv4 Network
  - Implement an IPv4 addressing scheme to enable end-to-end connectivity in a small to medium-sized business network.
  - Explain how subnetting segments a network to enable better communication.
  - Explain how to calculate IPv4 subnets for a /24 prefix.
  - Explain how to calculate IPv4 subnets for a /16 and /8 prefix.
  - Given a set of requirements for subnetting, implement an IPv4 addressing scheme.
  - Explain how to create a flexible addressing scheme using variable length subnet masking (VLSM).
- 8.2 Addressing Schemes
  - Given a set of requirements, implement a VLSM addressing scheme to provide connectivity to end users in a small to medium-sized network.
  - Implement a VLSM addressing scheme.

# Chapter 8 - Sections & Objectives (Cont.)

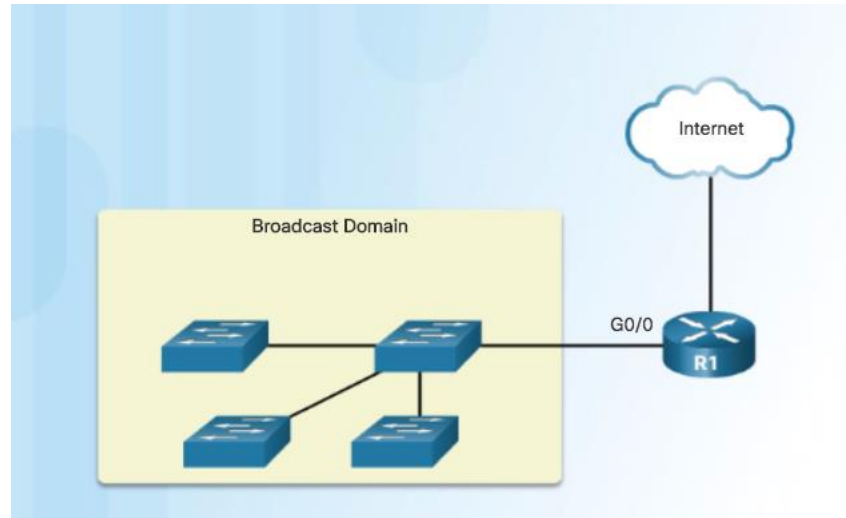
- 8.3 Address Schemes
  - Explain design considerations for implementing IPv6 in a business network.
  - Explain how to implement IPv6 address assignments in a business network.

# 8.1 Subnetting an IPv4 Network

## Network Segmentation

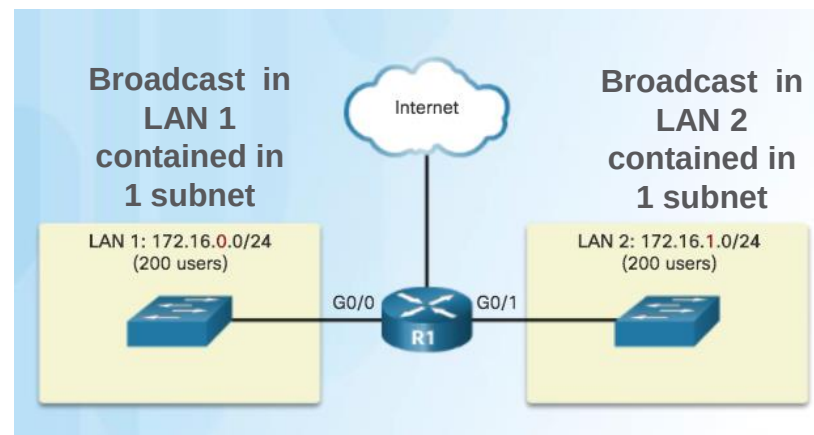
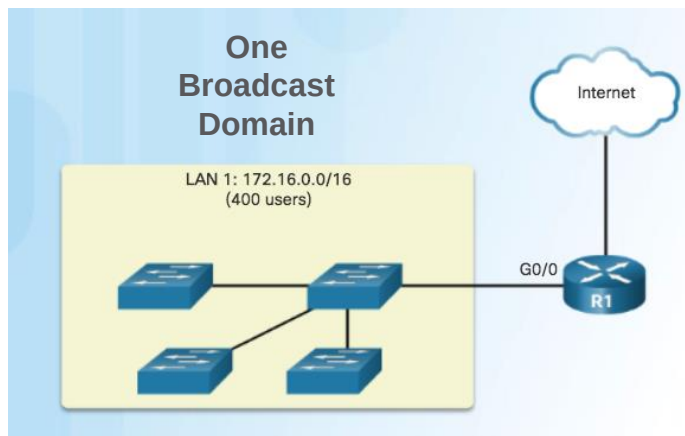
# Broadcast Domains

- Devices use broadcasts in an Ethernet LAN to locate:
  - **Other devices** - Address Resolution Protocol (ARP) which sends Layer 2 broadcasts to a known IPv4 address on the local network to discover the associated MAC address.
  - **Services** – Dynamic Host Configuration Protocol (DHCP) which sends broadcasts on the local network to locate a DHCP server.
- Switches propagate broadcasts out all interfaces except the interface on which it was received.



# Problems with Large Broadcast Domains

- Hosts can generate excessive broadcasts and negatively affect the network.
  - Slow network operations due to the significant amount of traffic it can cause.
  - Slow device operations because a device must accept and process each broadcast packet.
- Solution: Reduce the size of the network to create smaller broadcast domains. These smaller network spaces are called *subnets*.

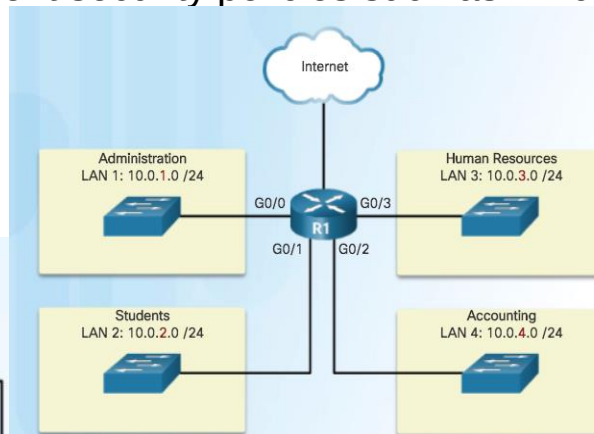
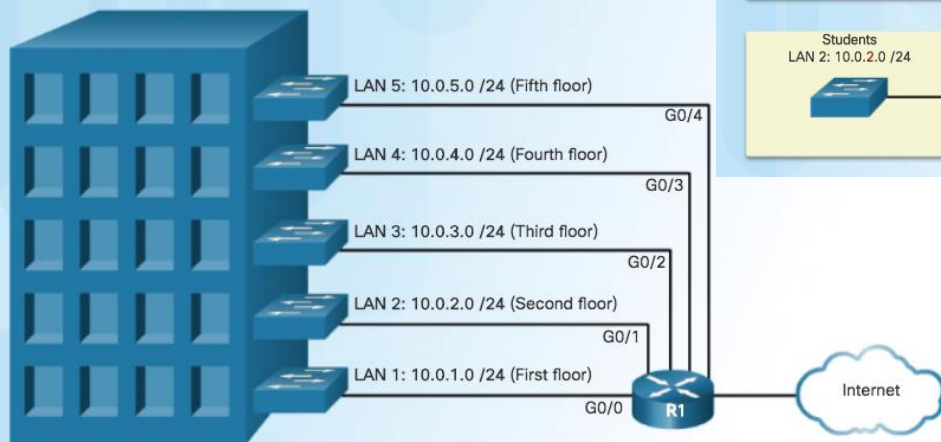


## Network Segmentation

# Reasons for Subnetting

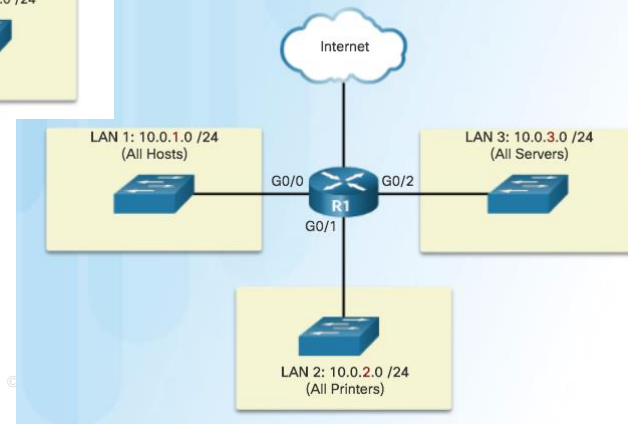
- Reduces overall network traffic and improves network performance.
- Enables an administrator to implement security policies such as which subnets are allowed or not allowed to communicate together.

### Subnetting by Location



### Communicating between Networks

### Subnetting by Device Type



# Subnetting an IPv4 Network

## Octet Boundaries

Networks are most easily subnetted at the octet boundary of /8, /16, and /24

| Prefix Length | Subnet Mask   | Subnet Mask in Binary<br>(n = network, h = host)                                                                 | # of hosts |
|---------------|---------------|------------------------------------------------------------------------------------------------------------------|------------|
| /8            | 255.0.0.0     | <code>nnnnnnnn . hhhhhhhh . hhhhhhhh . hhhhhhhh</code><br><code>11111111 . 00000000 . 00000000 . 00000000</code> | 16,777,214 |
| /16           | 255.255.0.0   | <code>nnnnnnnn . nnnnnnnn . hhhhhhhh . hhhhhhhh</code><br><code>11111111 . 11111111 . 00000000 . 00000000</code> | 65,534     |
| /24           | 255.255.255.0 | <code>nnnnnnnn . nnnnnnnn . nnnnnnnn . hhhhhhhh</code><br><code>11111111 . 11111111 . 11111111 . 00000000</code> | 254        |

- Prefix length and the subnet mask are different ways of identifying the network portion of an address.
- Subnets are created by borrowing host bits for network bits.
- More host bits borrowed, the more subnets that can be defined.



# Subnetting on the Octet Boundary

| Subnet Address<br>(256 Possible Subnets) | Host Range<br>(65,534 possible hosts per subnet) | Broadcast             |
|------------------------------------------|--------------------------------------------------|-----------------------|
| <u>10.0.0.0/16</u>                       | <u>10.0.0.1</u> – <u>10.0.255.254</u>            | <u>10.0.255.255</u>   |
| <u>10.1.0.0/16</u>                       | <u>10.1.0.1</u> – <u>10.1.255.254</u>            | <u>10.1.255.255</u>   |
| <u>10.2.0.0/16</u>                       | <u>10.2.0.1</u> – <u>10.2.255.254</u>            | <u>10.2.255.255</u>   |
| <u>10.3.0.0/16</u>                       | <u>10.3.0.1</u> – <u>10.3.255.254</u>            | <u>10.3.255.255</u>   |
| <u>10.4.0.0/16</u>                       | <u>10.4.0.1</u> – <u>10.4.255.254</u>            | <u>10.4.255.255</u>   |
| <u>10.5.0.0/16</u>                       | <u>10.5.0.1</u> – <u>10.5.255.254</u>            | <u>10.5.255.255</u>   |
| <u>10.6.0.0/16</u>                       | <u>10.6.0.1</u> – <u>10.6.255.254</u>            | <u>10.6.255.255</u>   |
| <u>10.7.0.0/16</u>                       | <u>10.7.0.1</u> – <u>10.7.255.254</u>            | <u>10.7.255.255</u>   |
| ...                                      | ...                                              | ...                   |
| <u>10.255.0.0/16</u>                     | <u>10.255.0.1</u> – <u>10.255.255.254</u>        | <u>10.255.255.255</u> |

- Subnetting Network 10.x.0.0/16
- Define up to 256 subnets with each subnet capable of connecting 65,534 hosts.
- First two octets identify the network portion while the last two octets are for host IP addresses.

# Subnetting on the Octet Boundary (Cont.)

| Subnet Address<br>(65,536 Possible Subnets) | Host Range<br>(254 possible hosts per subnet) | Broadcast             |
|---------------------------------------------|-----------------------------------------------|-----------------------|
| <u>10.0.0.0/24</u>                          | <u>10.0.0.1</u> - <u>10.0.0.254</u>           | <u>10.0.0.255</u>     |
| <u>10.0.1.0/24</u>                          | <u>10.0.1.1</u> - <u>10.0.1.254</u>           | <u>10.0.1.255</u>     |
| <u>10.0.2.0/24</u>                          | <u>10.0.2.1</u> - <u>10.0.2.254</u>           | <u>10.0.1.255</u>     |
| ...                                         | ...                                           | ...                   |
| <u>10.0.255.0/24</u>                        | <u>10.0.255.1</u> - <u>10.0.255.254</u>       | <u>10.0.255.255</u>   |
| <u>10.1.0.0/24</u>                          | <u>10.1.0.1</u> - <u>10.1.0.254</u>           | <u>10.1.0.255</u>     |
| <u>10.1.1.0/24</u>                          | <u>10.1.1.1</u> - <u>10.1.1.254</u>           | <u>1.1.1.0.255</u>    |
| <u>10.1.2.0/24</u>                          | <u>10.1.2.1</u> - <u>10.1.2.254</u>           | <u>10.1.2.0.255</u>   |
| ...                                         | ...                                           | ...                   |
| <u>10.100.0.0/24</u>                        | <u>10.100.0.1</u> - <u>10.100.0.254</u>       | <u>10.100.0.255</u>   |
| ...                                         | ...                                           | ...                   |
| <u>10.255.255.0/24</u>                      | <u>10.255.255.1</u> - <u>10.255.255.254</u>   | <u>10.255.255.255</u> |

- Subnetting Network 10.x.x.0/24
- Define 65,536 subnets each capable of connecting 254 hosts.
- /24 boundary is very popular in subnetting because of number of hosts.

# Subnetting an IPv4 Network

## Classless Subnetting

### Subnetting a /24 Network

| Prefix Length | Subnet Mask     | Subnet Mask in Binary<br>(n = network, h = host)                                                                                               | # of subnets | # of hosts |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------|
| /25           | 255.255.255.128 | n n n n n n n n . n n n n n n n n . n n n n n n n n . n h h h h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 0 0 0 0 0 0 0 | 2            | 126        |
| /26           | 255.255.255.192 | n n n n n n n n . n n n n n n n n . n n n n n n n n . n n h h h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 0 0 0 0 0 0 | 4            | 62         |
| /27           | 255.255.255.224 | n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n h h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0 | 8            | 30         |
| /28           | 255.255.255.240 | n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 0 0 0 0 | 16           | 14         |
| /29           | 255.255.255.248 | n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n n h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 0 0 0 | 32           | 6          |
| /30           | 255.255.255.252 | n n n n n n n n . n n n n n n n n . n n n n n n n n . n n n n n n h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0 | 64           | 2          |

Subnets can borrow bits from *any* host bit position to create other masks.

# Video Demonstration – The Subnet Mask

## Subnetting in Binary

### ▪ ANDING

- Convert IP address and Subnet Mask to Binary (line up vertically like an addition problem)
- Logically AND (1 and 1 = 1, all other combinations = 0)
- Result is network address for original IP address

### ▪ Classful Subnetting

- Class A /8 255.0.0.0
- Class B /16 255.255.0.0
- Class C /24 255.255.255.0



# Video Demonstration – The Subnet Mask (Cont.)

Subnetting 192.168.1.0/24

| 192      | 168      | 1        | 0        |
|----------|----------|----------|----------|
| 255      | 255      | 255      | 128      |
| 11000000 | 10101000 | 00000001 | 00000000 |
| 11111111 | 11111111 | 11111111 | 10000000 |
| N        | N        | N        | Sn H     |
|          |          |          |          |

Subnet bits =  $2^1 = 2$   
Host bits =  $2^7 = 128 - 2 = 126$   
Subnetworks = 2

Subnetting 192.168.1.0/24

| 192      | 168      | 1        | 68       |
|----------|----------|----------|----------|
| 255      | 255      | 255      | 128      |
| 11000000 | 10101000 | 00000001 | 01000100 |
| 11111111 | 11111111 | 11111111 | 10000000 |
| 11000000 | 10101000 | 00000001 | 00000000 |
| 192      | 168      | 1        | 0        |

192.168.1.0 /25 -----> 192.168.1.127 /25  
192.168.1.128 /25 -----> 192.168.1.255 /25

# Video Demonstration – Subnetting with the Magic Number

- Magic number technique used to calculate subnets
- Magic number is simply the place value of the last one in the subnet mask
- /25 11111111.11111111.11111111.**1**0000000 magic number = **128**
- /26 11111111.11111111.11111111.**11**000000 magic number = **64**
- /27 11111111.11111111.11111111.**111**00000 magic number = **32**



# Video Demonstration – Subnetting with the Magic Number (Cont.)

The Magic Number is the last 1 in Binary

|          |          |          |          |
|----------|----------|----------|----------|
| 192      | 168      | 1        | 0        |
| 255      | 255      | 255      | 224      |
| 11000000 | 10101000 | 00000001 | 00000000 |
| 11111111 | 11111111 | 11111111 | 11100000 |
|          |          |          | SN H     |

The Magic Number is? 32

- 192.168.1.0 /27    192.168.1.128 /27
- 192.168.1.32 /27    192.168.1.160 /27
- 192.168.1.64 /27    192.168.1.192 /27
- 192.168.1.96 /27    192.168.1.224 /27



# Video Demonstration – Subnetting with the Magic Number (Cont.)

Subnetting 172.16.0.0/16 -->/23

|          |          |           |          |
|----------|----------|-----------|----------|
| 172      | 16       | 0         | 0        |
| 255      | 255      | 254       | 0        |
| 10101010 | 00010000 | 00000000  | 00000000 |
| 11111111 | 11111111 | 1111 1110 | 00000000 |
|          |          | Sn        | H        |
|          |          |           |          |

What is the magic number? 2

172.16.0.0 ---- 172.16.1.255 /23

172.16.2.0 /23

172.16.4.0 /23



# Subnetting an IPv4 Network

## Classless Subnetting Example

**192.168.1.0/25 Network**

Borrow 1 bit from the host portion of the address.

Original

|      |      |    |   |          |
|------|------|----|---|----------|
| 192. | 168. | 1. | 0 | 000 0000 |
|------|------|----|---|----------|

Mask

|      |      |      |   |          |
|------|------|------|---|----------|
| 255. | 255. | 255. | 0 | 000 0000 |
|------|------|------|---|----------|

1 Network

The borrowed bit value is 0 for the Net 0 address.

Net 0

|      |      |    |   |          |
|------|------|----|---|----------|
| 192. | 168. | 1. | 0 | 000 0000 |
|------|------|----|---|----------|

Net 1

|      |      |    |   |          |
|------|------|----|---|----------|
| 192. | 168. | 1. | 1 | 000 0000 |
|------|------|----|---|----------|

2 Subnets

The new subnets have the SAME subnet mask.

Mask

|      |      |      |   |          |
|------|------|------|---|----------|
| 255. | 255. | 255. | 1 | 000 0000 |
|------|------|------|---|----------|

**Dotted Decimal Addresses**

Borrow 1 bit from the host portion of the address.

Original

|      |      |    |   |          |
|------|------|----|---|----------|
| 192. | 168. | 1. | 0 | 000 0000 |
|------|------|----|---|----------|

Mask

|      |      |      |   |          |
|------|------|------|---|----------|
| 255. | 255. | 255. | 0 | 000 0000 |
|------|------|------|---|----------|

1 Network

192. 168. 1. **0/25**

Net 0

|      |      |    |   |          |
|------|------|----|---|----------|
| 192. | 168. | 1. | 0 | 000 0000 |
|------|------|----|---|----------|

192. 168. 1. **128/25**

Net 1

|      |      |    |   |          |
|------|------|----|---|----------|
| 192. | 168. | 1. | 1 | 000 0000 |
|------|------|----|---|----------|

2 Subnets

255. 255. 255. **128**

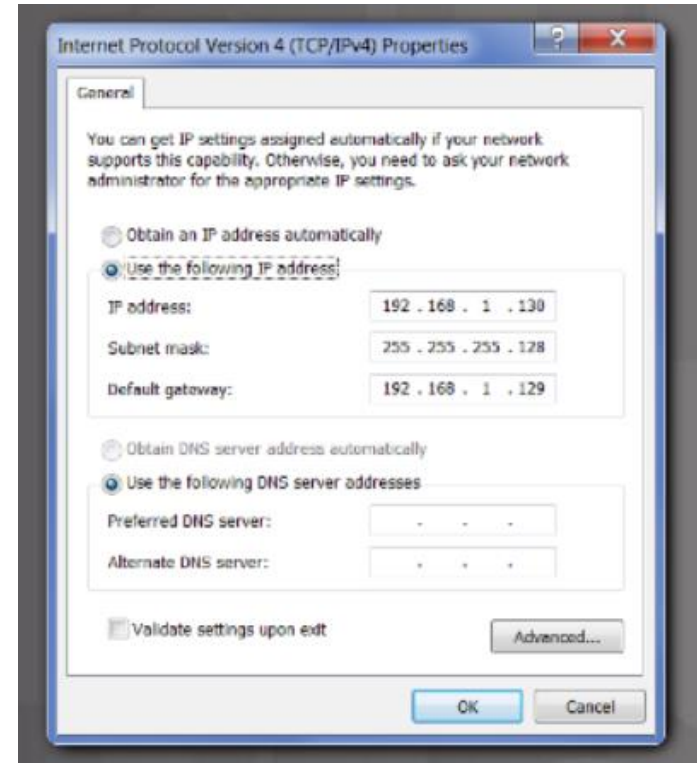
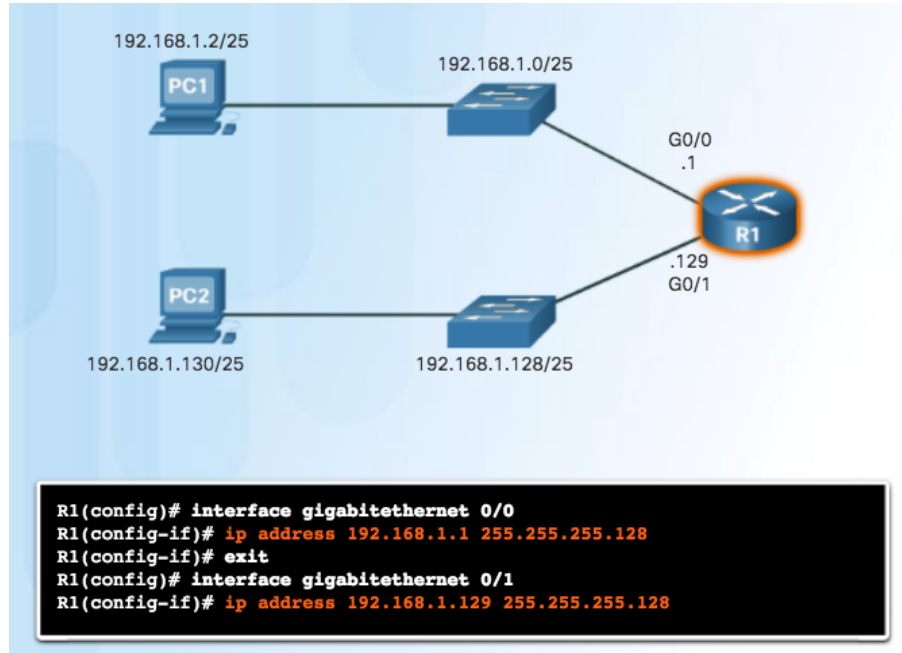
Mask

|      |      |      |   |          |
|------|------|------|---|----------|
| 255. | 255. | 255. | 1 | 000 0000 |
|------|------|------|---|----------|

# Subnetting an IPv4 Network

## Creating 2 Subnets

### ■ /25 Subnetting Topology



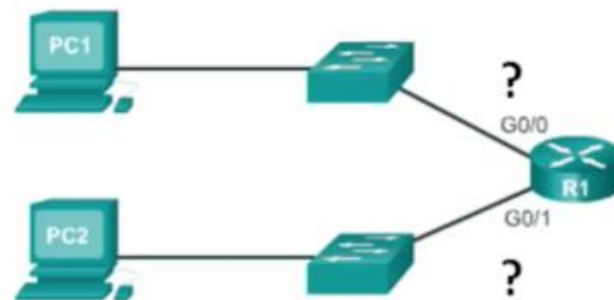
# Video Demonstration – Creating Two Equal-sized Subnets (/25)

Create 2 Equal-sized Subnets from 192.168.1.0 /24

- **Subnet Mask** - 11111111.11111111.11111111.**1**0000000

| $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 128   | 64    | 32    | 16    | 8     | 4     | 2     | 1     |
| 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

- Magic Number = **128**
- 192.168.1.0 /25 (**start at 0**)
- 192.168.1.128 /25 (**Add 128**)



# Subnetting an IPv4 Network

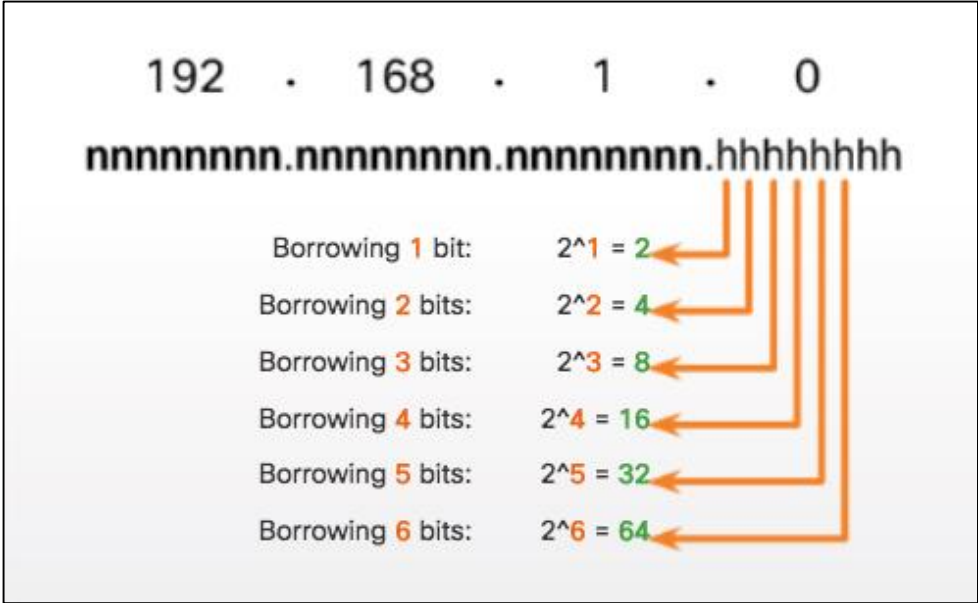
## Subnetting Formulas

Calculate Number of Subnets Formula

$$2^n$$

*n* = bits borrowed

Subnetting a /24 Network



# Subnetting an IPv4 Network

## Subnetting Formulas (Cont.)

Calculate Number of  
Hosts Formula

$$2^n - 2$$

*n* = the number of bits remaining in the host  
field

Calculating the Number of Hosts

192. 168. 1. 0 000 0000

7 bits remain in host field

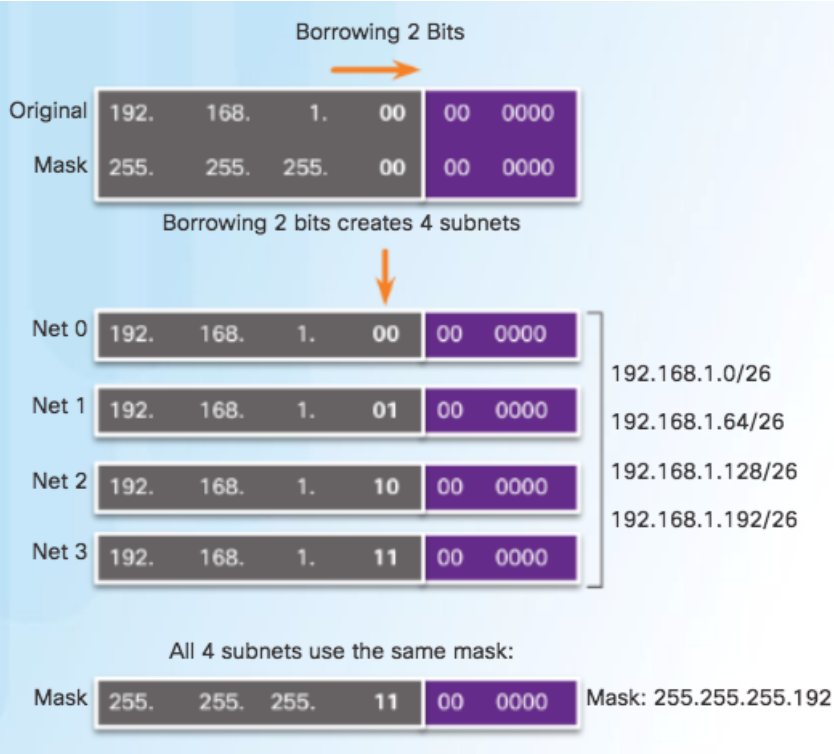
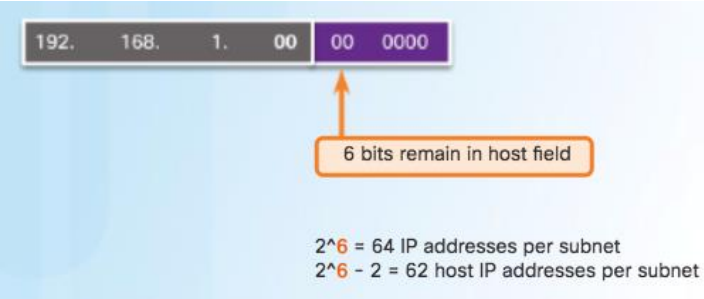
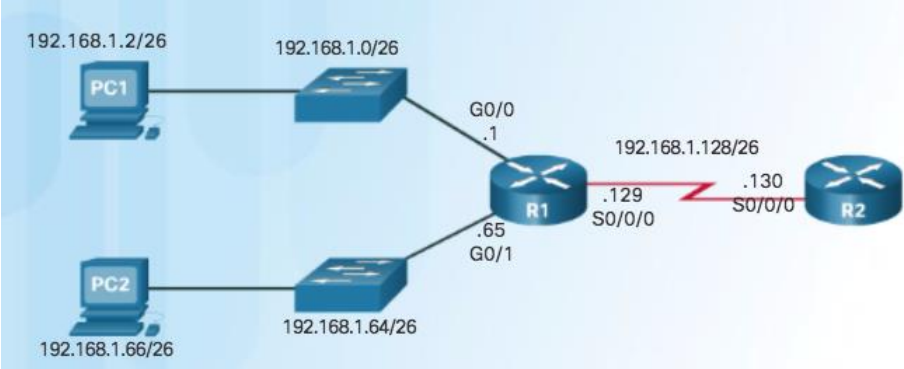
$2^7 = 128$  IP addresses per subnet

$2^7 - 2 = 126$  host IP addresses per subnet

# Subnetting an IPv4 Network

## Creating 4 Subnets

- /26 Subnetting Topology



# Subnetting an IPv4 Network

## Creating 4 Subnets (Cont.)

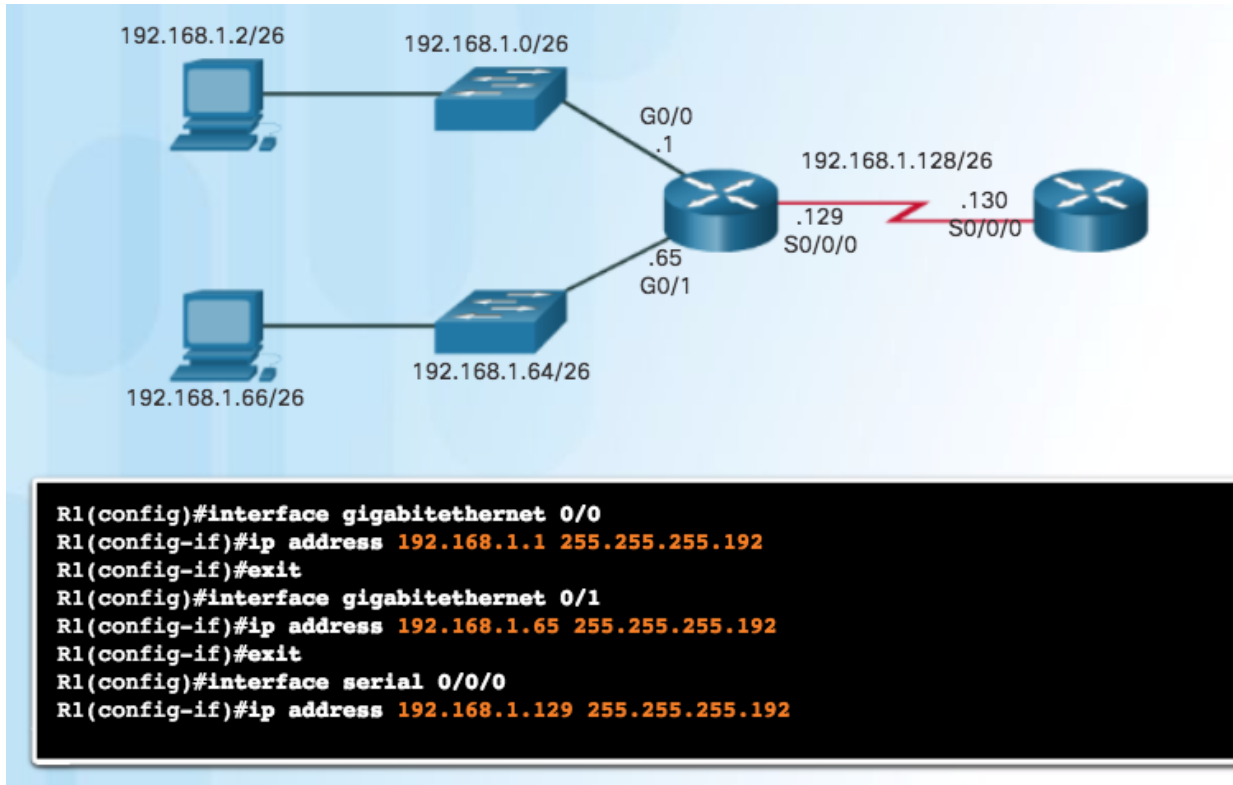
- /26 Subnetting Topology

|       |           |      |      |    |    |    |      |               |
|-------|-----------|------|------|----|----|----|------|---------------|
| Net 0 | Network   | 192. | 168. | 1. | 00 | 00 | 0000 | 192.168.1.0   |
|       | First     | 192. | 168. | 1. | 00 | 00 | 0001 | 192.168.1.1   |
|       | Last      | 192. | 168. | 1. | 00 | 11 | 1110 | 192.168.1.62  |
|       | Broadcast | 192. | 168. | 1. | 00 | 11 | 1111 | 192.168.1.63  |
| Net 1 | Network   | 192. | 168. | 1. | 01 | 00 | 0000 | 192.168.1.64  |
|       | First     | 192. | 168. | 1. | 01 | 00 | 0001 | 192.168.1.65  |
|       | Last      | 192. | 168. | 1. | 01 | 11 | 1110 | 192.168.1.126 |
|       | Broadcast | 192. | 168. | 1. | 01 | 11 | 1111 | 192.168.1.127 |
| Net 2 | Network   | 192. | 168. | 1. | 10 | 00 | 0000 | 192.168.1.128 |
|       | First     | 192. | 168. | 1. | 10 | 00 | 0001 | 192.168.1.129 |
|       | Last      | 192. | 168. | 1. | 10 | 11 | 1110 | 192.168.1.190 |
|       | Broadcast | 192. | 168. | 1. | 10 | 11 | 1111 | 192.168.1.191 |

# Subnetting an IPv4 Network

## Creating 4 Subnets (Cont.)

- /26 Subnetting Topology





# Video Demonstration – Creating Four Equal-sized Subnets (/26)

Create 4 Equal-sized Subnets from 192.168.1.0 /24

- Subnet Mask in Binary – 11111111.11111111.11111111.**11**000000
- $2^2 = 4$  Subnets
- Magic Number = 64
- 192.168.1.0 /26
- 192.168.1.64 /26
- 192.168.1.128 /26
- 192.168.1.192 /26



# Video Demonstration – Creating Eight Equal-sized Subnets (/27)

Create 8 Equal-sized Subnets from 192.168.1.0 /24

- Borrow 3 bits – 11111111.11111111.11111111.**111**00000
- Magic Number = 32
- 192.168.1.0 /27 (Start at 0)
- 192.168.1.32 /27 (Add 32 to previous network)
- 192.168.1.64 /27 (Add 32)
- 192.168.1.96 /27 (Add 32)
- 192.168.1.128 /27 (Add 32)
- 192.168.1.160 /27 (Add 32)
- 192.168.1.192 /27 (Add 32)
- 192.168.1.224 /27 (Add 32)



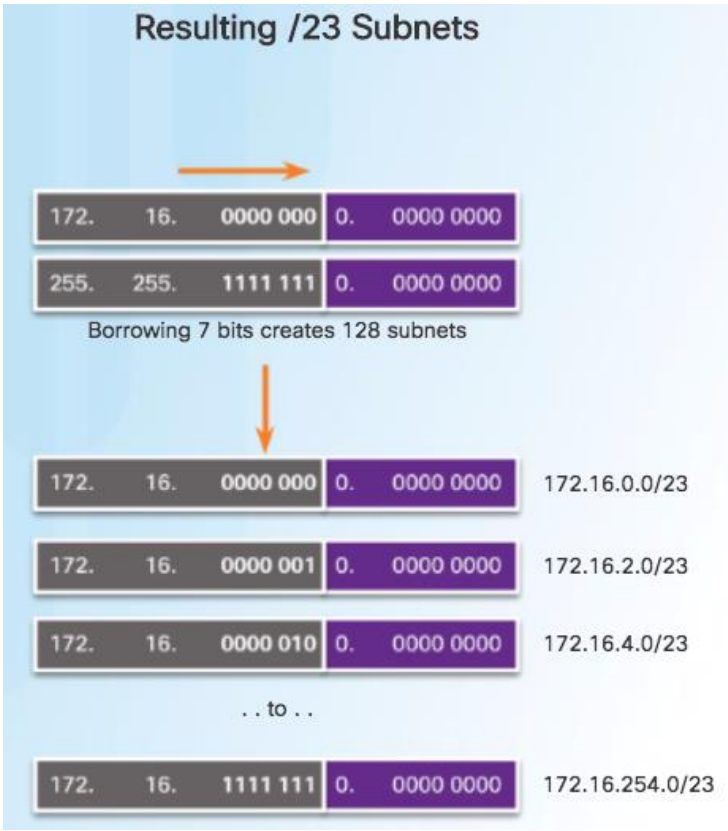
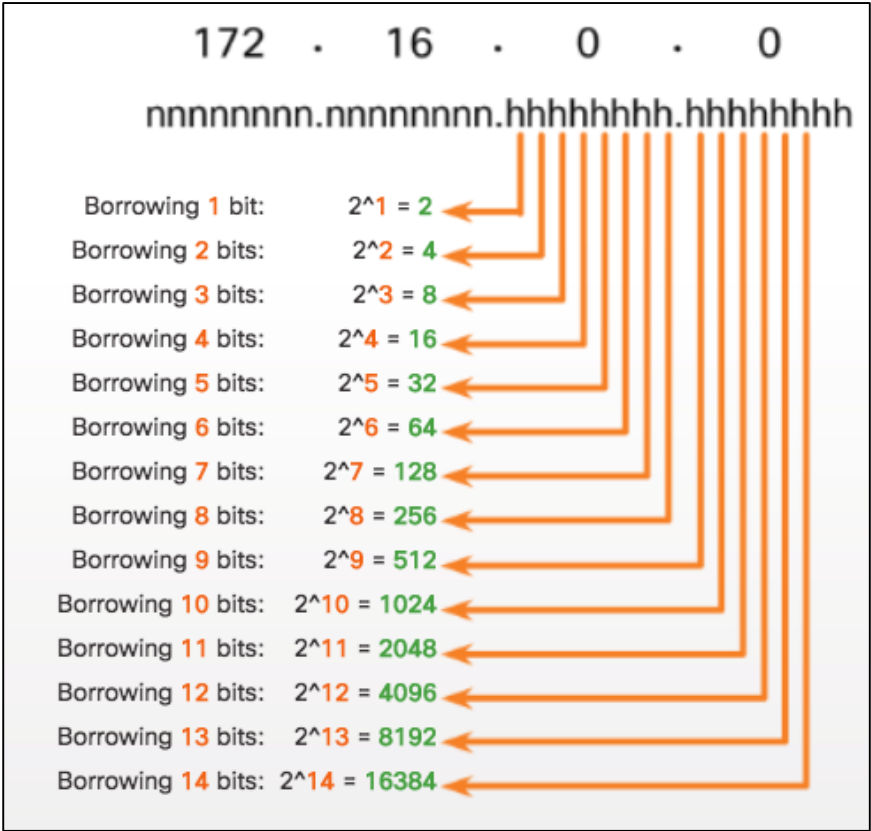
Subnetting a /16 and /8 Prefix

# Creating Subnets with a /16 prefix

| Subnetting a /16 Network |               |                                                                                |              |            |
|--------------------------|---------------|--------------------------------------------------------------------------------|--------------|------------|
| Prefix Length            | Subnet Mask   | Network Address<br>(n = network, h = host)                                     | # of subnets | # of hosts |
| /17                      | 255.255.128.0 | nnnnnnnn.nnnnnnnn.nhshshshh.hshshshshh<br>11111111.11111111.10000000.00000000  | 2            | 32766      |
| /18                      | 255.255.192.0 | nnnnnnnn.nnnnnnnn.nnshshshh.hshshshshh<br>11111111.11111111.11000000.00000000  | 4            | 16382      |
| /19                      | 255.255.224.0 | nnnnnnnn.nnnnnnnn.nnnshshshh.hshshshshh<br>11111111.11111111.11100000.00000000 | 8            | 8190       |
| /20                      | 255.255.240.0 | nnnnnnnn.nnnnnnnn.nnnnshshh.hshshshshh<br>11111111.11111111.11110000.00000000  | 16           | 4094       |
| /21                      | 255.255.248.0 | nnnnnnnn.nnnnnnnn.nnnnnshh.hshshshshh<br>11111111.11111111.11111000.00000000   | 32           | 2046       |
| /22                      | 255.255.252.0 | nnnnnnnn.nnnnnnnn.nnnnnnsh.hshshshshh<br>11111111.11111111.11111100.00000000   | 64           | 1022       |

# Subnetting a /16 and /8 Prefix

## Creating 100 Subnets with a /16 prefix



# Subnetting a /16 and /8 Prefix

## Calculating the Hosts

Hosts =  $2^n$   
(where n = host bits remaining)

|      |     |            |              |
|------|-----|------------|--------------|
| 172. | 16. | 00 00 00 0 | 0. 0000 0000 |
|------|-----|------------|--------------|

9 bits remain in host field

$2^9 = 512$  IP addresses per subnet  
 $2^9 - 2 = 510$  host IP addresses per subnet

### Address Range for 172.16.0.0/23 Subnet

Network Address

|      |     |            |              |
|------|-----|------------|--------------|
| 172. | 16. | 00 00 00 0 | 0. 0000 0000 |
|------|-----|------------|--------------|

 = 172.16.0.0/23

First Host Address

|      |     |            |              |
|------|-----|------------|--------------|
| 172. | 16. | 00 00 00 0 | 0. 0000 0001 |
|------|-----|------------|--------------|

 = 172.16.0.1/23

Last Host Address

|      |     |            |              |
|------|-----|------------|--------------|
| 172. | 16. | 00 00 00 0 | 1. 1111 1110 |
|------|-----|------------|--------------|

 = 172.16.1.254/23

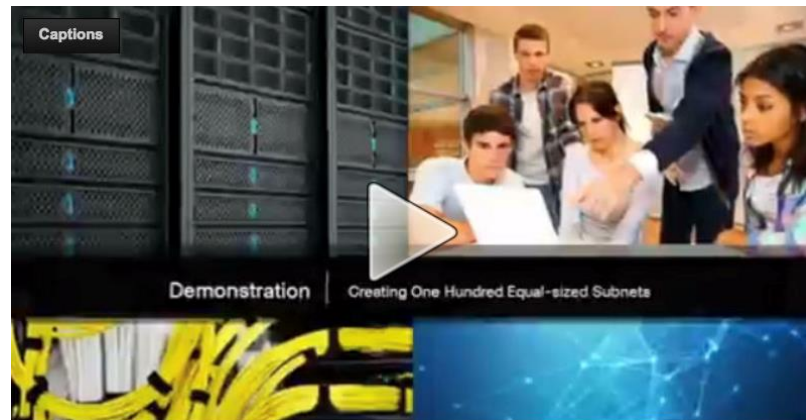
Broadcast Address

|      |     |            |              |
|------|-----|------------|--------------|
| 172. | 16. | 00 00 00 0 | 1. 1111 1111 |
|------|-----|------------|--------------|

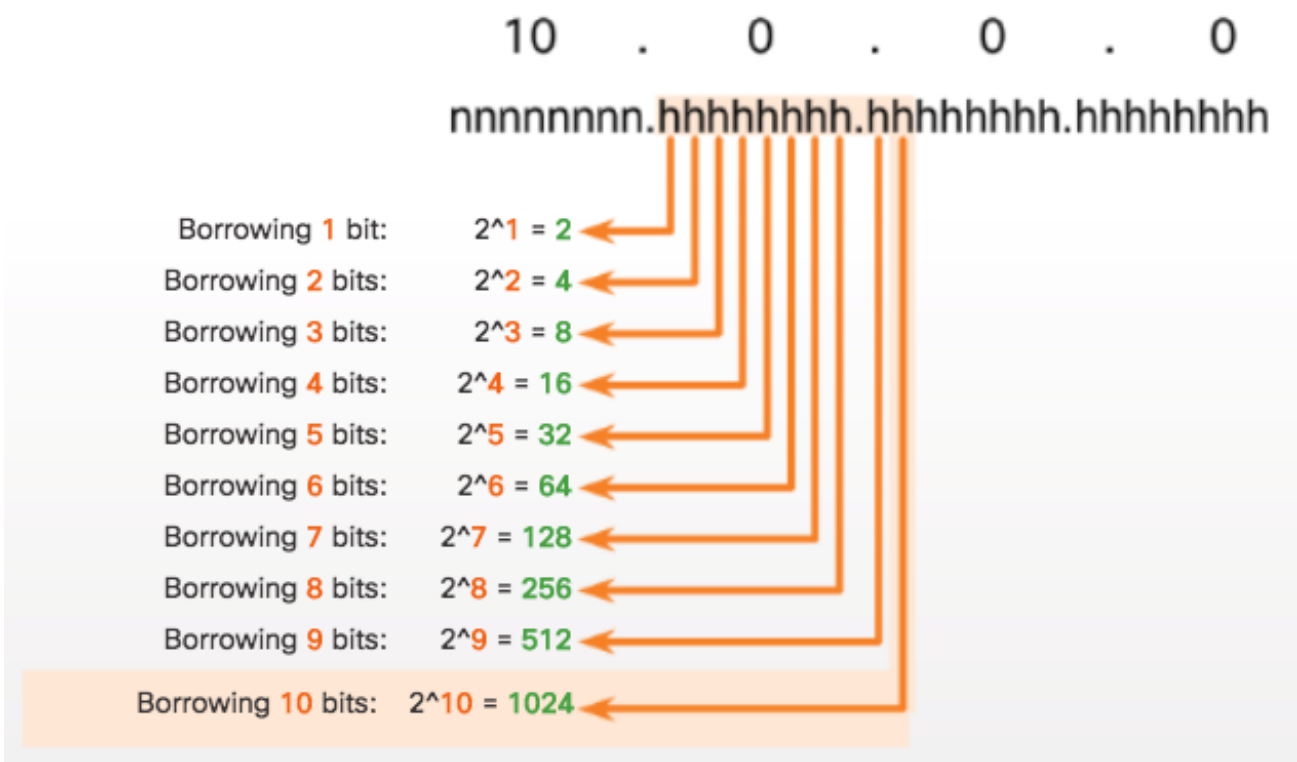
 = 172.16.1.255/23

# Video Demonstration – Creating One Hundred Equal-sized Subnets

- An enterprise network requires 100 equal-sized subnets starting from 172.16.0.0/16
  - New Subnet Mask
    - 11111111.11111111.**11111111**0.00000000
  - $2^7 = 128$  Subnets
  - $2^9 = 512$  hosts per subnet
  - Magic Number = 2
  - 172.16.**0**.0 /23
  - 172.16.**2**.0 /23
  - 172.16.**4**.0 /23
  - 172.16.**6**.0 /23
  - ...
  - 172.16.**254**.0 /23

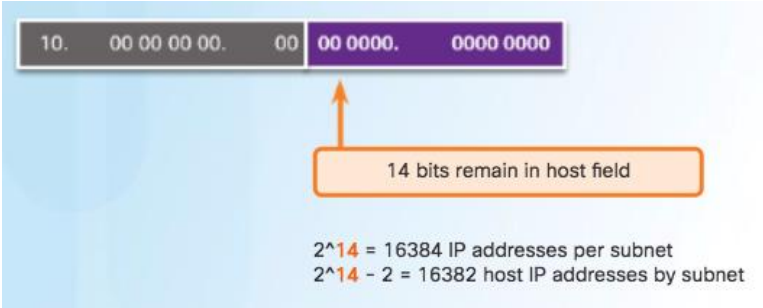
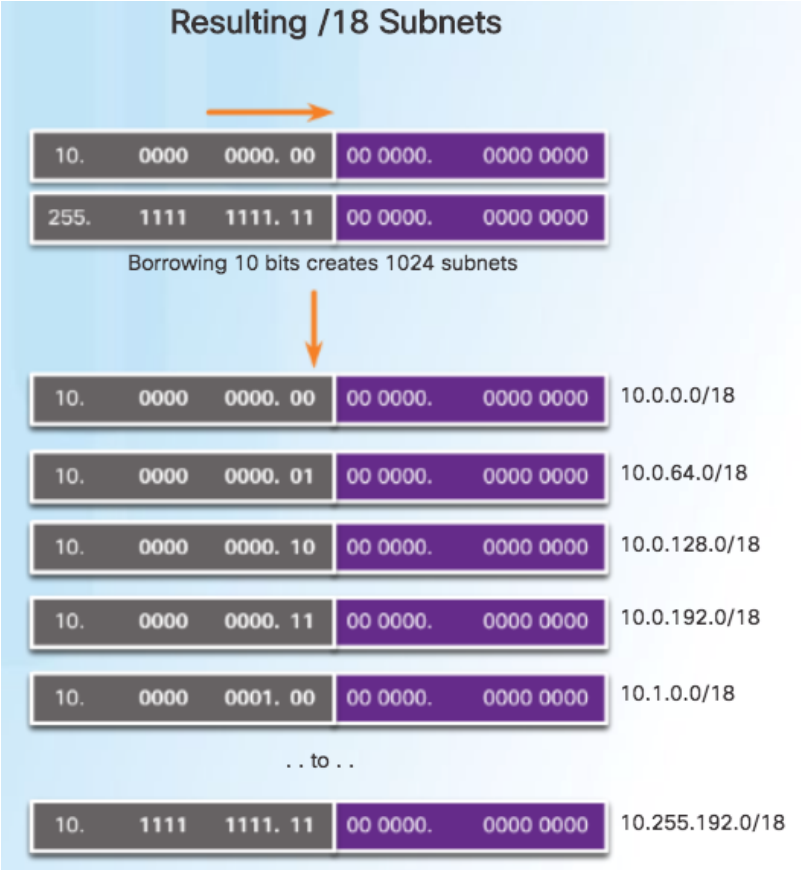


# Creating 1000 Subnets with a /8 Network



Subnetting a /16 and /8 Prefix

Creating 1000 Subnets with a /8 Network (Cont.)





# Video Demonstration – Subnetting Across Multiple Octets

The Magic Number is the last 1 in Binary

|          |          |          |          |
|----------|----------|----------|----------|
| 10       | 0        | 0        | 0        |
| 255      | 0        | 0        | 0        |
| 00001010 | 00000000 | 00000000 | 00000000 |
| 11111111 | 11100000 | 00000000 | 00000000 |
|          | SN       | H        | H        |

The Magic Number is? 32

10.0.0.0 /11

10.32.0.0 /11

10.64.0.0 /11

10.96.0.0 /11

10.128.0.0 /11

10.160.0.0 /11

10.192.0.0 – 10.223.255.255 /11

10.224.0.0 /11



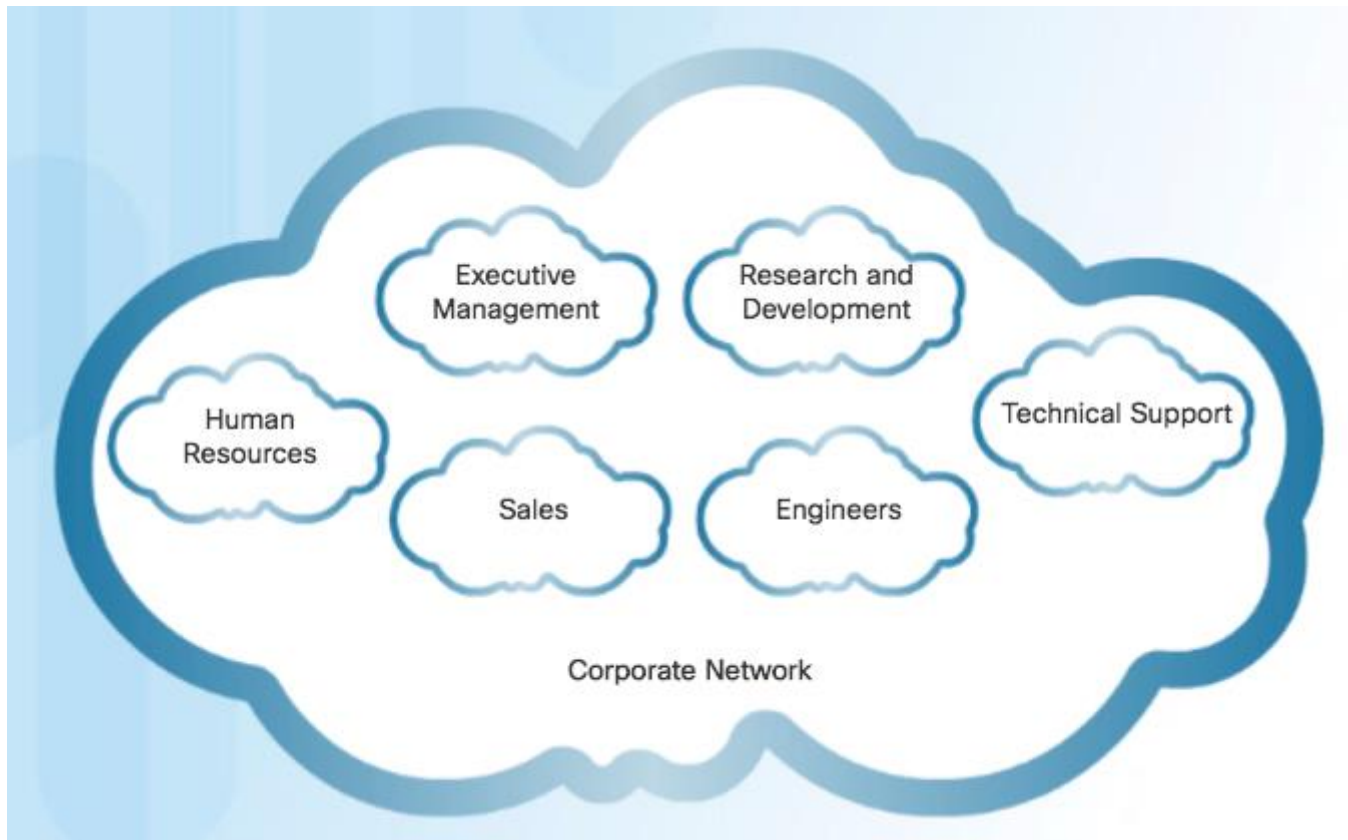
New Challenge Problem: Create over 300 Equal-sized Subnets of 20,000 Hosts each starting from 10.0.0.0/8

# Subnetting Based on Host Requirements

| Prefix Length | Subnet Mask     | Subnet Mask in Binary<br>(n = network, h = host)                                                                             | # of subnets | # of hosts |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------|--------------|------------|
| /25           | 255.255.255.128 | n n n n n n n n . n n n n n n n n . n h h h h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 0 0 0 0 0 0 0 | 2            | 126        |
| /26           | 255.255.255.192 | n n n n n n n n . n n n n n n n n . n n h h h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 0 0 0 0 0 0 | 4            | 62         |
| /27           | 255.255.255.224 | n n n n n n n n . n n n n n n n n . n n n h h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0 | 8            | 30         |
| /28           | 255.255.255.240 | n n n n n n n n . n n n n n n n n . n n n n h h h h<br>1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 0 0 0 0 | 16           | 14         |

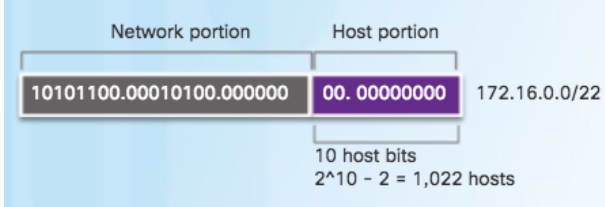
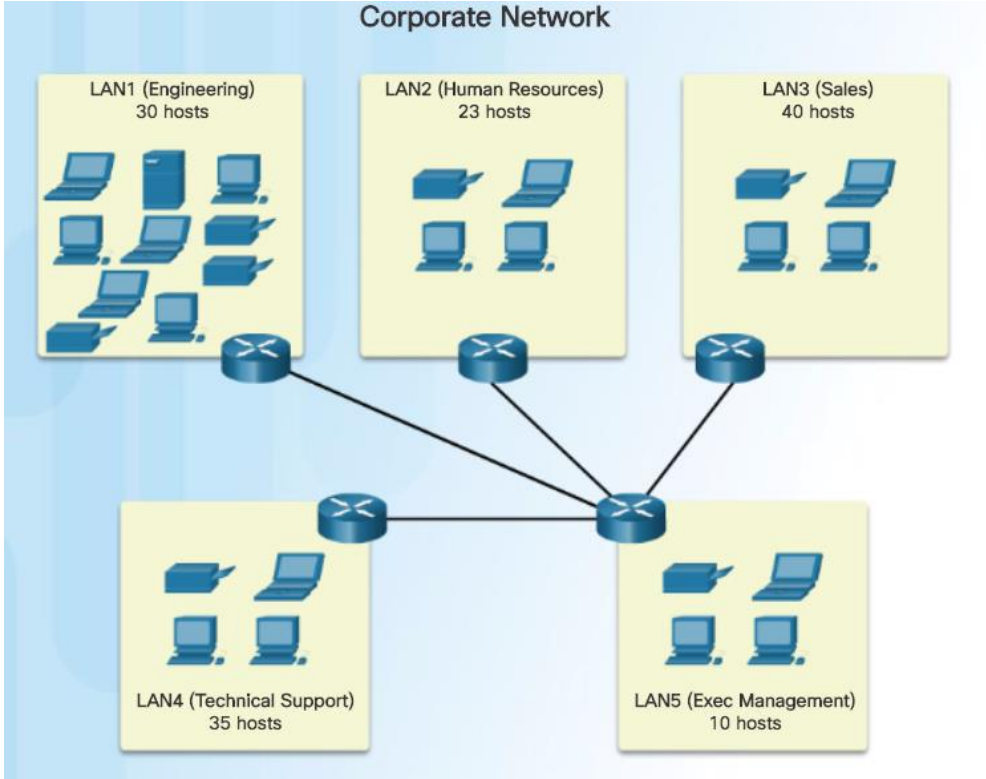
# Subnetting Based On Network Requirements

Host devices used by employees in the Engineering department in one network and Management in a separate network.



# Subnetting to Meet Requirements

## Network Requirement Example

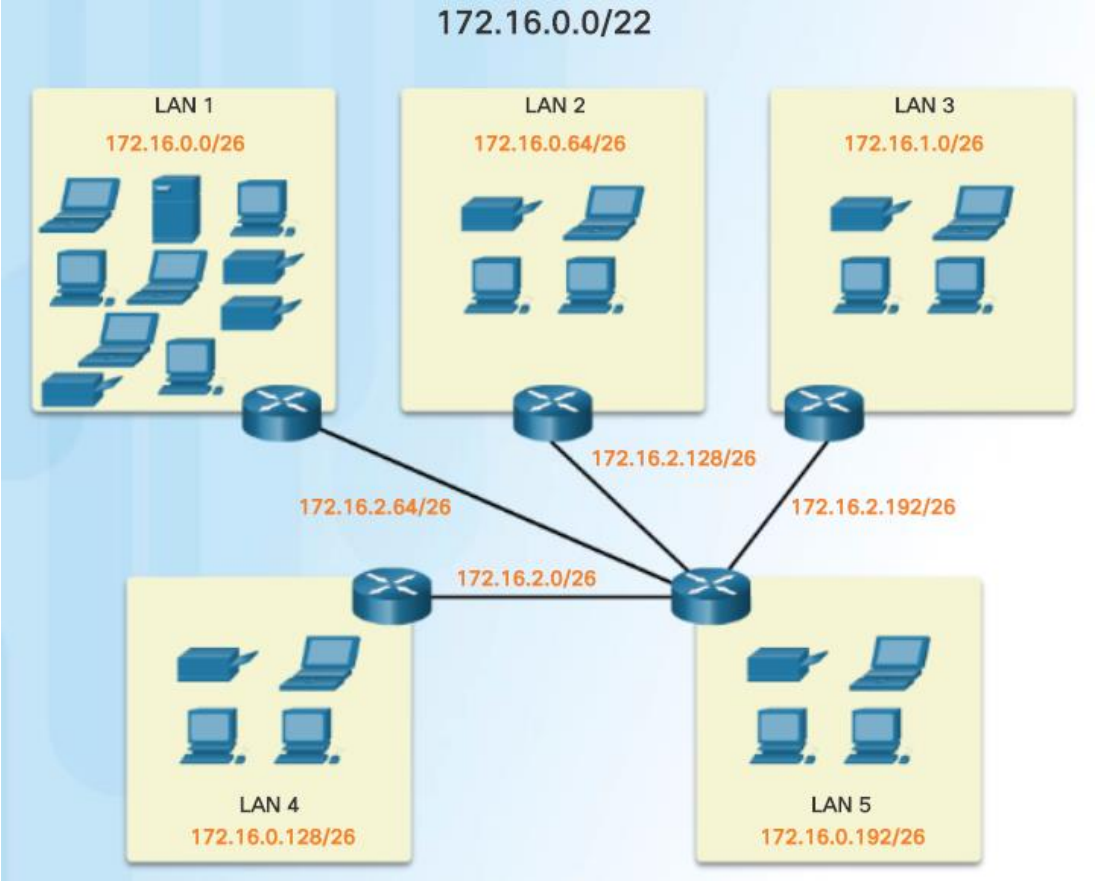


|                       | Network Portion          | Host Portion |        | Dotted Decimal  |
|-----------------------|--------------------------|--------------|--------|-----------------|
|                       | 10101100.00010000.000000 | 00.00        | 000000 | 172.16.0.0/22   |
| 0                     | 10101100.00010000.000000 | 00.00        | 000000 | 172.16.0.0/26   |
| 1                     | 10101100.00010000.000000 | 00.01        | 000000 | 172.16.0.64/26  |
| 2                     | 10101100.00010000.000000 | 00.10        | 000000 | 172.16.0.128/26 |
| 3                     | 10101100.00010000.000000 | 00.11        | 000000 | 172.16.0.192/26 |
| 4                     | 10101100.00010000.000000 | 01.00        | 000000 | 172.16.1.0/26   |
| 5                     | 10101100.00010000.000000 | 01.01        | 000000 | 172.16.1.64/26  |
| 6                     | 10101100.00010000.000000 | 01.10        | 000000 | 172.16.1.128/26 |
| Nets 7 - 13 not shown |                          |              |        |                 |
| 14                    | 10101100.00010000.000000 | 11.10        | 000000 | 172.16.3.128/26 |
| 15                    | 10101100.00010000.000000 | 11.11        | 000000 | 172.16.3.192/26 |

4 bits borrowed from host portion to create subnets

# Subnetting to Meet Requirements

## Network Requirement Example (Cont.)



# Lab – Calculating IPv4 Subnets



## Lab – Calculating IPv4 Subnets

### Objectives

**Part 1: Determine IPv4 Address Subnetting**

**Part 2: Calculate IPv4 Address Subnetting**

### Background / Scenario

The ability to work with IPv4 subnets and determine network and host information based on a given IP address and subnet mask is critical to understanding how IPv4 networks operate. The first part is designed to reinforce how to compute network IP address information from a given IP address and subnet mask. When given an IP address and subnet mask, you will be able to determine other information about the subnet.

### Required Resources

- 1 PC (Windows 7 or 8 with Internet access)
- Optional: IPv4 address calculator

### Part 1: Determine IPv4 Address Subnetting

In Part 1, you will determine the network and broadcast addresses, as well as the number of hosts, given an IPv4 address and subnet mask.

**REVIEW:** To determine the network address, perform binary ANDing on the IPv4 address using the subnet mask provided. The result will be the network address. Hint: If the subnet mask has decimal value 255 in an octet, the result will ALWAYS be the original value of that octet. If the subnet mask has decimal value 0 in an octet, the result will ALWAYS be 0 for that octet.

Example:

|                         |               |
|-------------------------|---------------|
| <b>IP Address</b>       | 192.168.10.10 |
| <b>Subnet Mask</b>      | 255.255.255.0 |
|                         | =====         |
| <b>Result (Network)</b> | 192.168.10.0  |

# Subnetting to Meet Requirements

## Packet Tracer – Subnetting Scenario

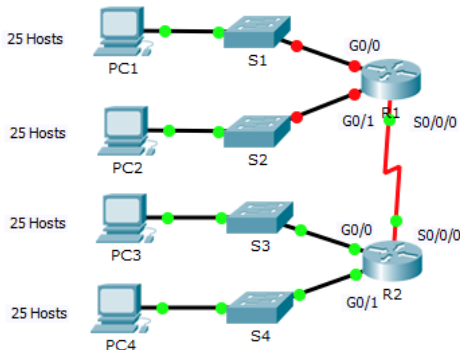


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### Packet Tracer - Subnetting Scenario

#### Topology



#### Addressing Table

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|--------|-----------|------------|-------------|-----------------|
| R1     | G0/0      |            |             |                 |
|        | G0/1      |            |             |                 |
|        | S0/0/0    |            |             |                 |
|        |           |            |             |                 |

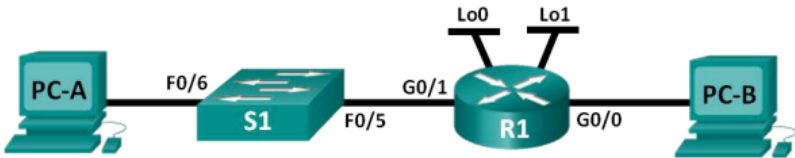
# Subnetting to Meet Requirements

## Lab – Designing and Implementing a Subnetted IPv4 Addressing Scheme



### Lab - Designing and Implementing a Subnetted IPv4 Addressing Scheme

#### Topology



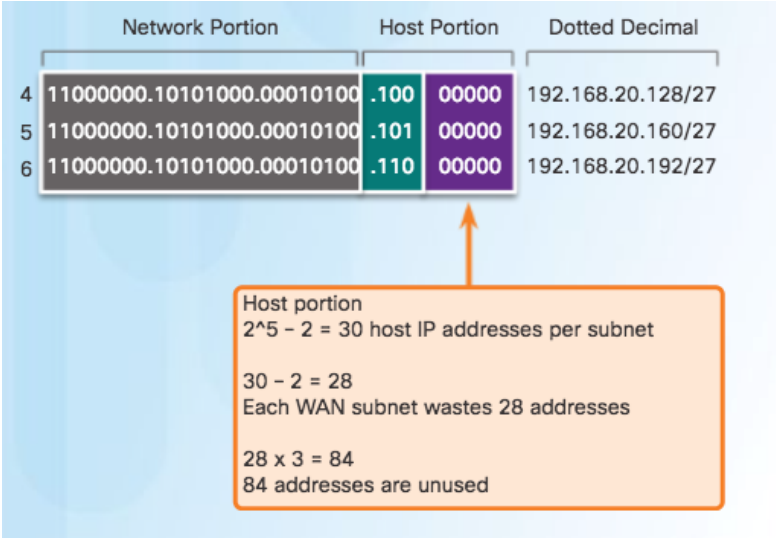
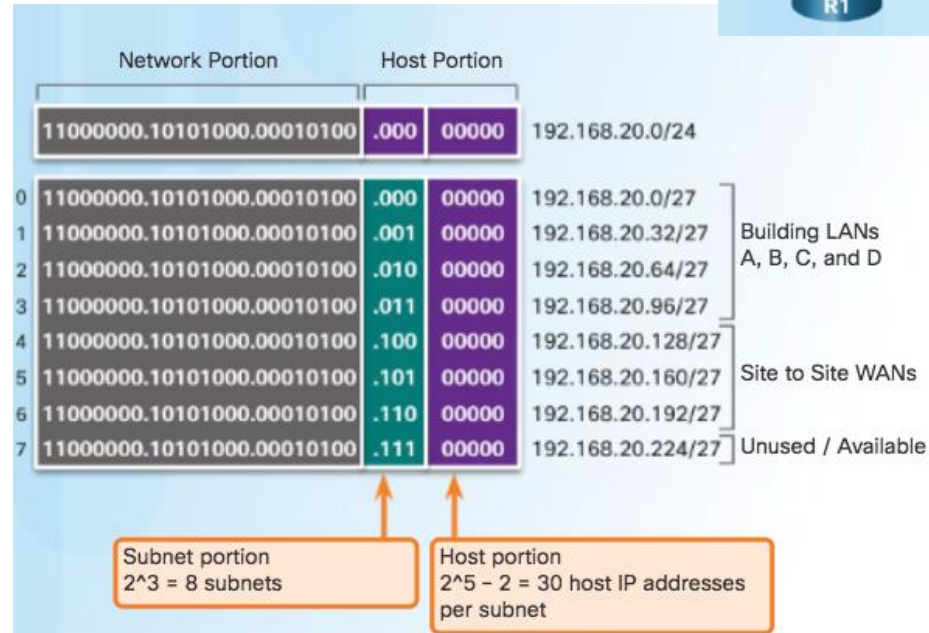
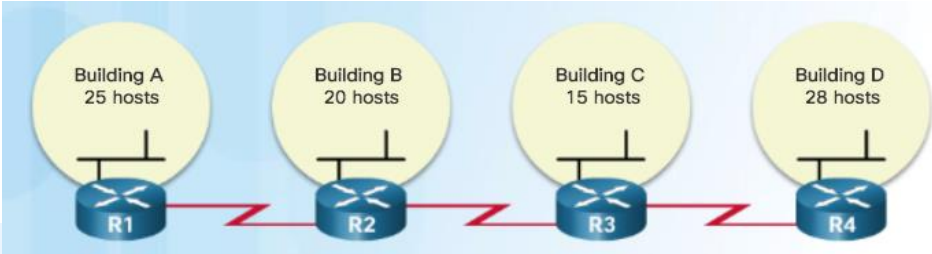
#### Addressing Table

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|--------|-----------|------------|-------------|-----------------|
| R1     | G0/0      |            |             | N/A             |
|        | G0/1      |            |             | N/A             |
|        | Lo0       |            |             | N/A             |
|        | Lo1       |            |             | N/A             |
| S1     | VLAN 1    | N/A        | N/A         | N/A             |
| PC-A   | NIC       |            |             |                 |
| PC-B   | NIC       |            |             |                 |



# Benefits of Variable Length Subnet Masking

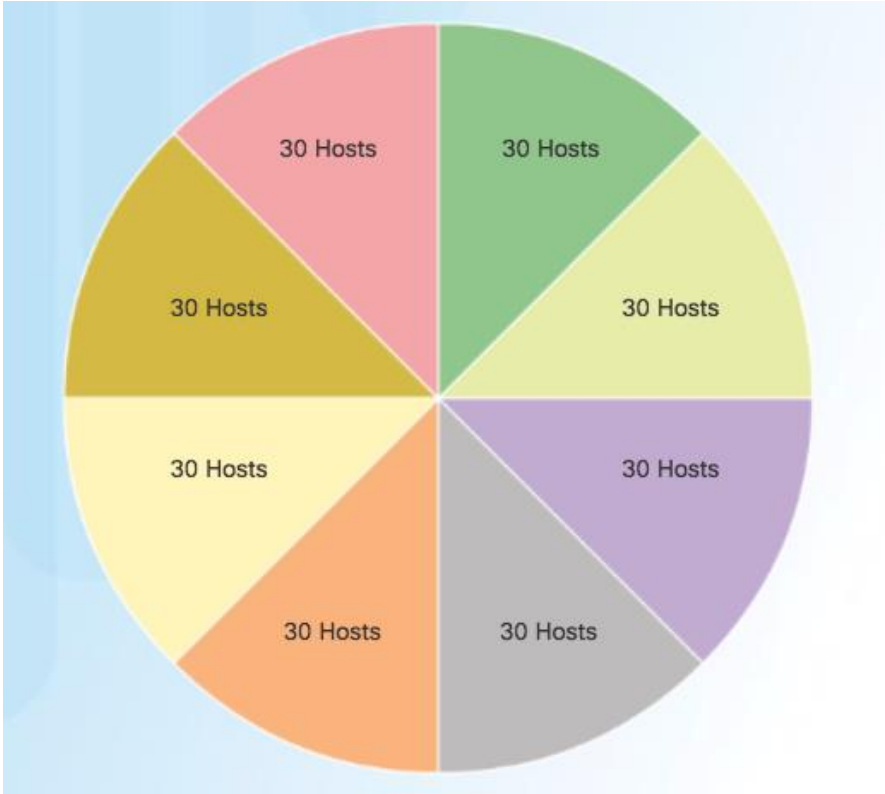
## Traditional Subnetting Wastes Addresses



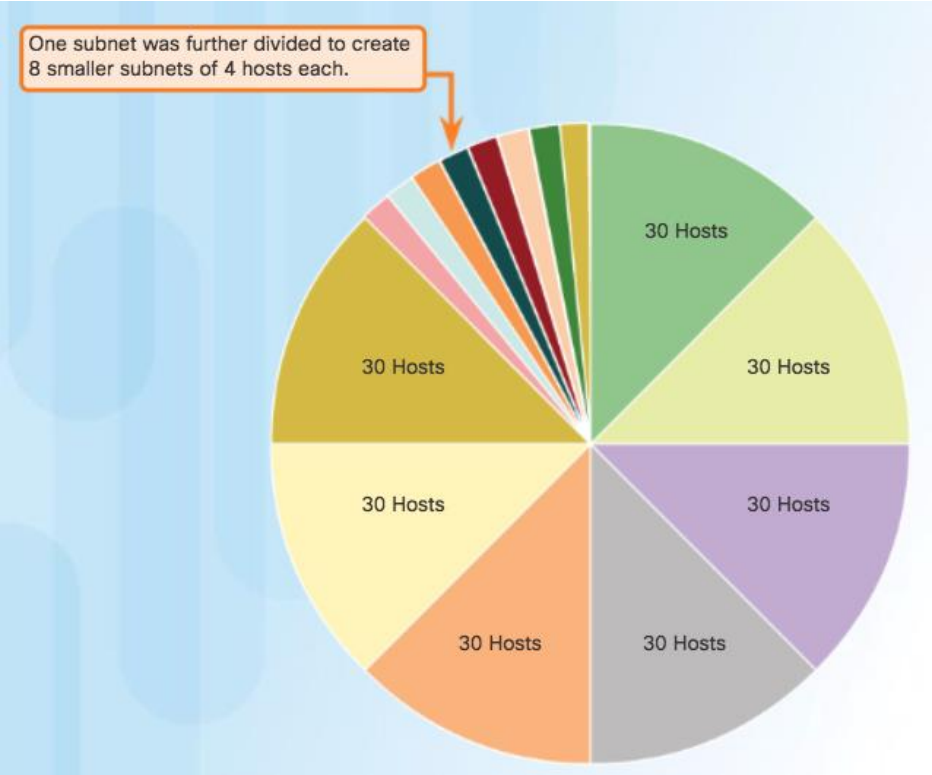
# Benefits of Variable Length Subnet Masking

## Variable Length Subnet Masks (VLSM)

Traditional



Subnets of Varying Sizes



# Benefits of Variable Length Subnet Masking

## Basic VLSM

### Basic Subnetting

|   | Network portion            | Host portion | Dotted Decimal  |                   |
|---|----------------------------|--------------|-----------------|-------------------|
|   | 11000000.10101000.00010100 | .00000000    | 192.168.20.0/24 |                   |
| 0 | 11000000.10101000.00010100 | .000         | 00000           | 192.168.20.0/27   |
| 1 | 11000000.10101000.00010100 | .001         | 00000           | 192.168.20.32/27  |
| 2 | 11000000.10101000.00010100 | .010         | 00000           | 192.168.20.64/27  |
| 3 | 11000000.10101000.00010100 | .011         | 00000           | 192.168.20.96/27  |
| 4 | 11000000.10101000.00010100 | .100         | 00000           | 192.168.20.128/27 |
| 5 | 11000000.10101000.00010100 | .101         | 00000           | 192.168.20.160/27 |
| 6 | 11000000.10101000.00010100 | .110         | 00000           | 192.168.20.192/27 |
| 7 | 11000000.10101000.00010100 | .111         | 00000           | 192.168.20.224/27 |

Subnet 7 will be subnetted further.

|                                    | Network portion            | Host portion | Dotted Decimal |                   |
|------------------------------------|----------------------------|--------------|----------------|-------------------|
| 7                                  | 11000000.10101000.00010100 | .111         | 00000          | 192.168.20.224/27 |
| 3 more bits borrowed from subnet 7 |                            |              |                |                   |
| 7:0                                | 11000000.10101000.00010100 | .111000      | 00             | 192.168.20.224/30 |
| 7:1                                | 11000000.10101000.00010100 | .111001      | 00             | 192.168.20.228/30 |
| 7:2                                | 11000000.10101000.00010100 | .111010      | 00             | 192.168.20.232/30 |
| 7:3                                | 11000000.10101000.00010100 | .111011      | 00             | 192.168.20.236/30 |
| 7:4                                | 11000000.10101000.00010100 | .111100      | 00             | 192.168.20.240/30 |
| 7:5                                | 11000000.10101000.00010100 | .111101      | 00             | 192.168.20.244/30 |
| 7:6                                | 11000000.10101000.00010100 | .111110      | 00             | 192.168.20.248/30 |
| 7:7                                | 11000000.10101000.00010100 | .111111      | 00             | 192.168.20.252/30 |

Subnetting a subnet

LANs  
A, B, C, D

Unused /  
Available

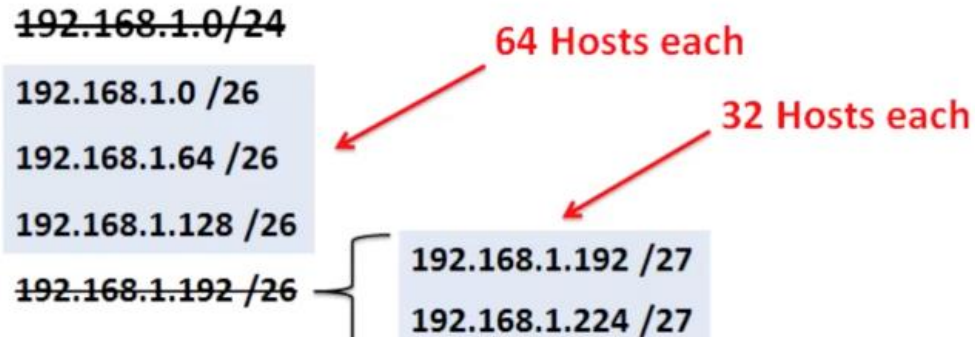
WANs

Unused /  
Available

# Benefits of Variable Length Subnet Masking

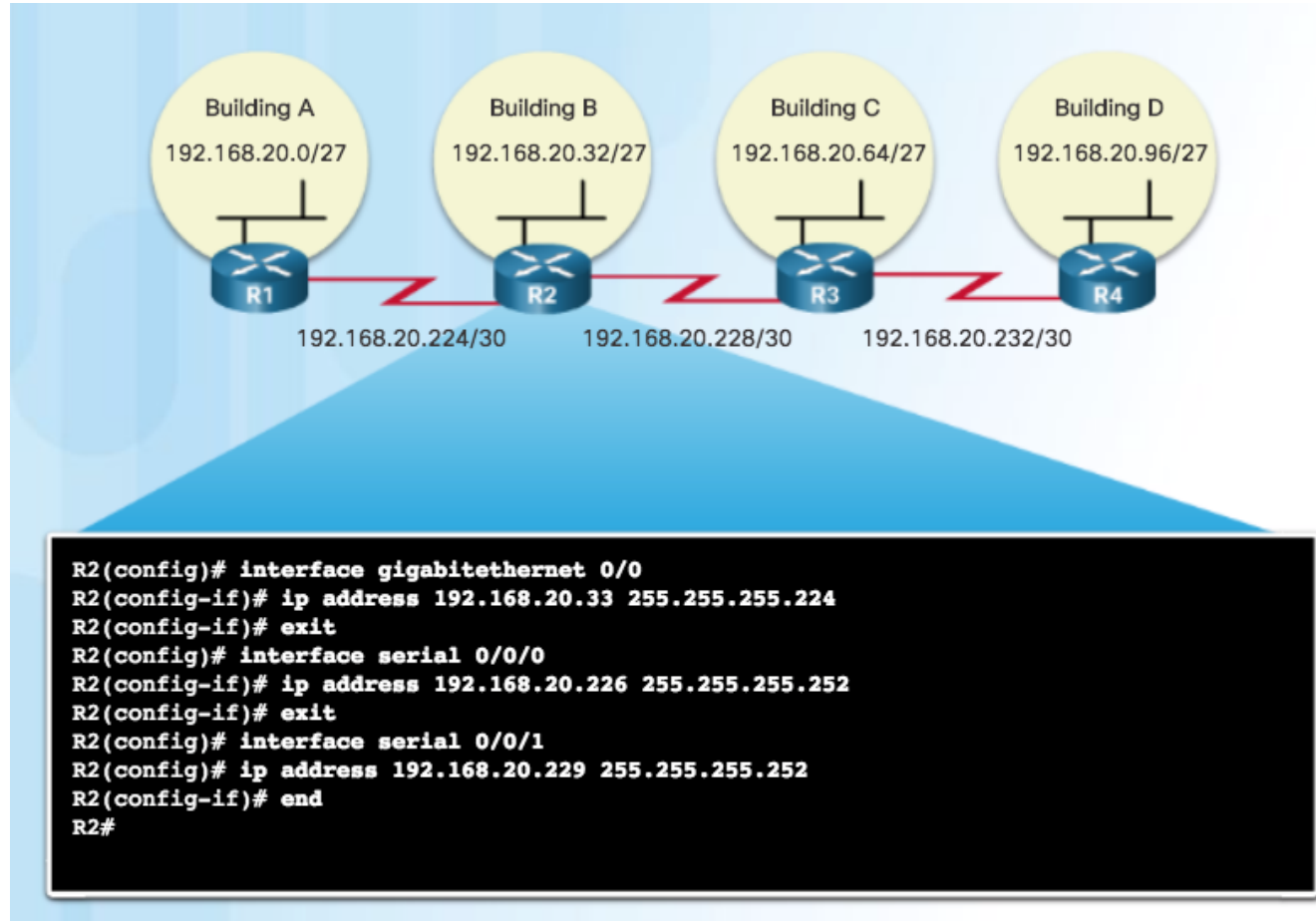
## Video Demonstration – VLSM Basics

- Basic VLSM
  - Subnets do not have to be equal sizes, as long as their address ranges do not overlap.
  - When creating subnets it is easier to work from larger to smaller.



# Benefits of Variable Length Subnet Masking

## VLSM in Practice



# Benefits of Variable Length Subnet Masking

## VLSM Chart

### VLSM Subnetting of 192.168.20.0/24

|        | /27 Network | Hosts       |
|--------|-------------|-------------|
| Bldg A | .0          | .1 - .30    |
| Bldg B | .32         | .33 - .62   |
| Bldg C | .64         | .65 - .94   |
| Bldg D | .96         | .97 - .126  |
| Unused | .128        | .129 - .158 |
| Unused | .160        | .161 - .190 |
| Unused | .192        | .193 - .222 |
|        | .224        | .225 - .254 |

|           | /30 Network | Hosts       |
|-----------|-------------|-------------|
| WAN R1-R2 | .224        | .225 - .226 |
| WAN R2-R3 | .228        | .229 - .230 |
| WAN R3-R4 | .232        | .233 - .234 |
| Unused    | .236        | .237 - .238 |
| Unused    | .240        | .241 - .242 |
| Unused    | .244        | .245 - .246 |
| Unused    | .248        | .249 - .250 |
| Unused    | .252        | .253 - .254 |



# Video Demonstration – VLSM Example

- Given the network 172.16.0.0  
/23 creates subnets:

- 1 network for 200 hosts - 256
- 1 network for 100 hosts - 128
- 1 network for 50 hosts - 64
- 1 network for 25 hosts - 32
- 1 network for 10 hosts - 16
- 4 point-to-point networks for 2 hosts each –  $4 \times 4 = 16$

/23 =  $2^9$  hosts = 512

$256 + 128 + 64 + 32 + 16 + 16 = 512$  hosts needed

Address range 172.16.0.0 – 172.16.1.255

~~172.16.0.0 /23~~

172.16.0.0 /24 (256)

~~172.16.1.0 /24~~

172.16.1.0 /25 (128)

~~172.16.1.128 /25~~

172.16.1.128 /26 (64)

~~172.16.1.192 /26~~

172.16.1.192 /27 (32)

~~172.16.1.224 /27~~

172.16.1.224 /28 (16)

~~172.16.1.240 /28~~

172.16.1.240 /30 (4)

172.16.1.244 /30 (4)

172.16.1.248 /30 (4)

172.16.1.252 /30 (4)



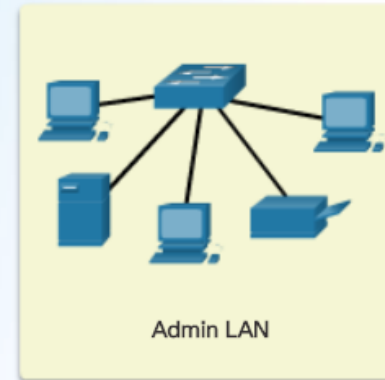
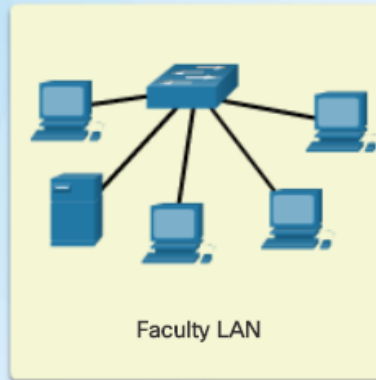
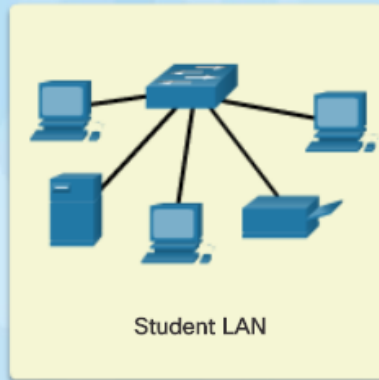
## 8.2 Addressing Schemes



# Structured Design

## Network Address Planning

### Planning IP Address Assignment



Planning requires decisions on each subnet in terms of size, the number of hosts per subnet, and how host addresses will be assigned.

# Planning to Address the Network



- Each host in an internetwork must have a unique address.
- Need proper planning & documentation.
- Must provide & control access to servers from internal hosts and external hosts.
- Layer 3 STATIC address assigned to a server can be used to control access to that server.
- Monitoring security and performance of hosts means network traffic is examined for source IP addresses that are generating or receiving excessive packets.

# Assigning Addresses to Devices

- Devices that require addresses:
  - **End user clients**
    - Can be set for DHCP to save time and manual errors.
    - A change in the subnetting scheme requires reconfiguration of DHCP server. IPv6 clients use DHCPv6/SLAAC.
  - **Servers**
    - Configured with static addresses.
    - Private addresses translated to public addresses if accessible from the Internet.
  - **Intermediary devices**
    - Set with static addresses for remote management.
  - **Gateway**
    - Router interface used to exit the network.

| Network: 192.168.1.0/24        |       |      |
|--------------------------------|-------|------|
| Use                            | First | Last |
| Host Devices                   | .1    | .229 |
| Servers                        | .230  | .239 |
| Printers                       | .240  | .249 |
| Intermediary Devices           | .250  | .253 |
| Gateway (router LAN interface) | .254  |      |

# Structured Design

## Packet Tracer – Designing and Implementing a VLSM Addressing Scheme



### Packet Tracer - Designing and Implementing a VLSM Addressing Scheme

#### Topology

You will receive one of three possible topologies.

#### Addressing Table

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|--------|-----------|------------|-------------|-----------------|
|        | G0/0      |            |             | N/A             |
|        | G0/1      |            |             | N/A             |
|        | S0/0/0    |            |             | N/A             |
|        | G0/0      |            |             | N/A             |
|        | G0/1      |            |             | N/A             |
|        | S0/0/0    |            |             | N/A             |
|        | VLAN 1    |            |             |                 |
|        | VLAN 1    |            |             |                 |
|        | VLAN 1    |            |             |                 |
|        | VLAN 1    |            |             |                 |
|        | NIC       |            |             |                 |
|        | NIC       |            |             |                 |
|        | NIC       |            |             |                 |
|        | NIC       |            |             |                 |

#### Objectives

Part 1: Examine the Network Requirements

Part 2: Design the VLSM Addressing Scheme

Part 3: Assign IP Addresses to Devices and Verify Connectivity

# Structured Design

## Lab – Designing and Implementing a VLSM Addressing Scheme

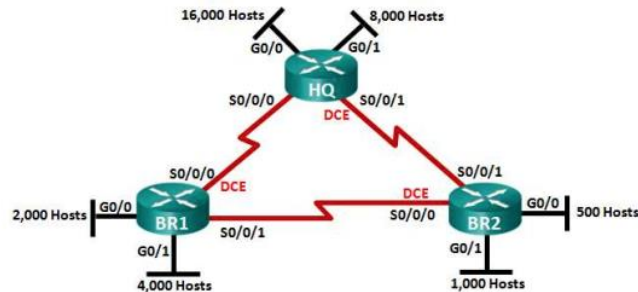


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### Lab – Designing and Implementing a VLSM Addressing Scheme

#### Topology



#### Objectives

Part 1: Examine Network Requirements

Part 2: Design the VLSM Address Scheme

Part 3: Cable and Configure the IPv4 Network

#### Background / Scenario

Variable Length Subnet Mask (VLSM) was designed to avoid wasting IP addresses. With VLSM, a network is subnetted and then re-subnetted. This process can be repeated multiple times to create subnets of various sizes based on the number of hosts required in each subnet. Effective use of VLSM requires address planning.

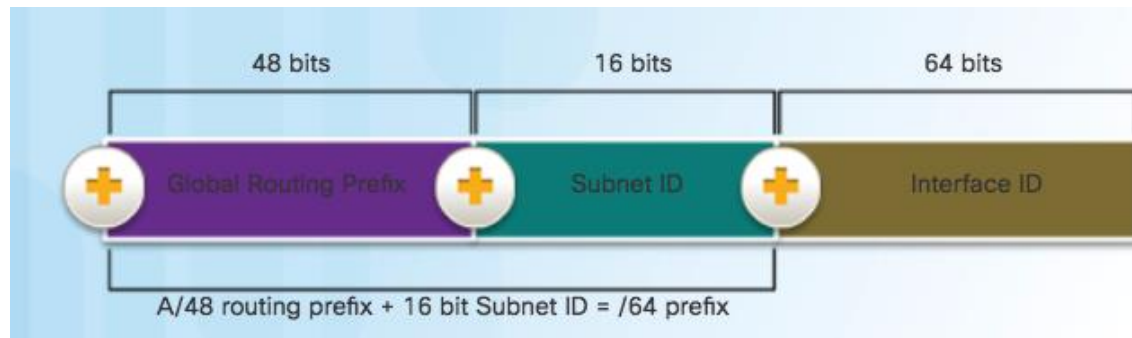
In this lab, use the 172.16.128.0/17 network address to develop an address scheme for the network displayed in the topology diagram. VLSM is used to meet the IPv4 addressing requirements. After you have designed the VLSM address scheme, you will configure the interfaces on the routers with the appropriate IP address information.

## 8.3 Design Considerations for IPv6

# The IPv6 Global Unicast Address

- IPv6 subnetting is not concerned with conserving address space.
- IPv6 subnetting is about building an addressing hierarchy based on the number of subnetworks needed.
- IPv6 link-local address is never subnetted.
- IPv6 global unicast address can be subnetted.
- IPv6 global unicast address normally consists of a /48 global routing prefix, a 16 bit subnet ID, and a 64 bit interface ID.

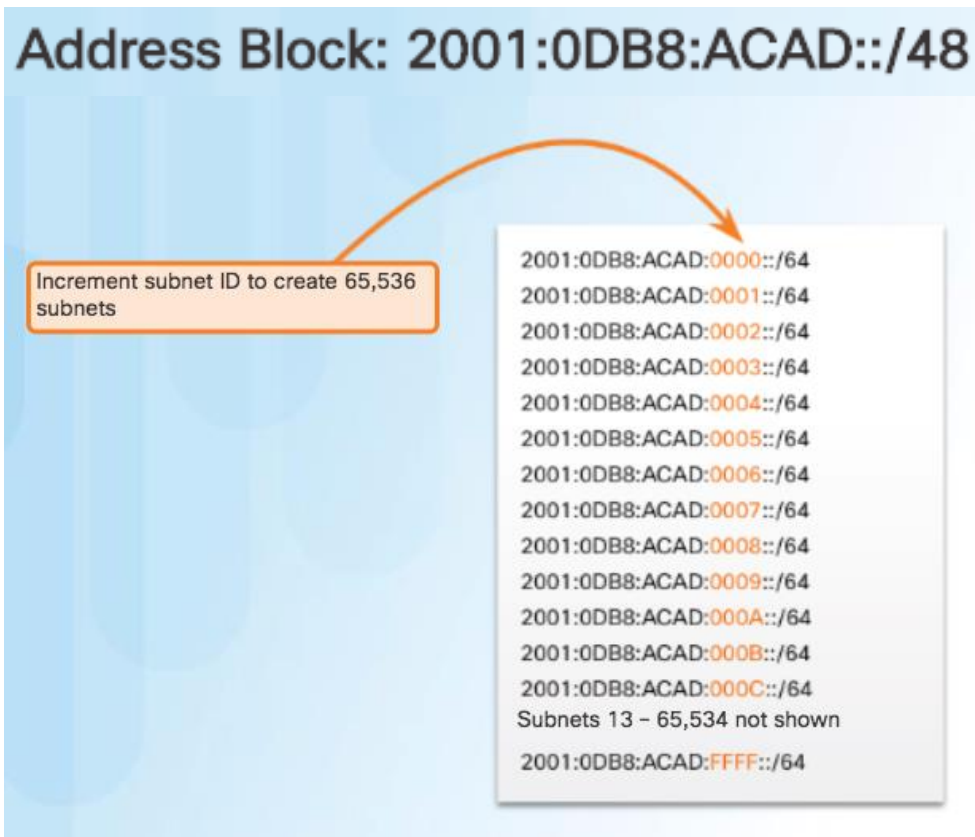
## Structure



### Global Routing Prefix

This is the prefix, or network, portion of the address that is assigned by the provider. Typically, Regional Internet Registries (RIRs) assign a /48 global routing prefix to ISPs and customers.

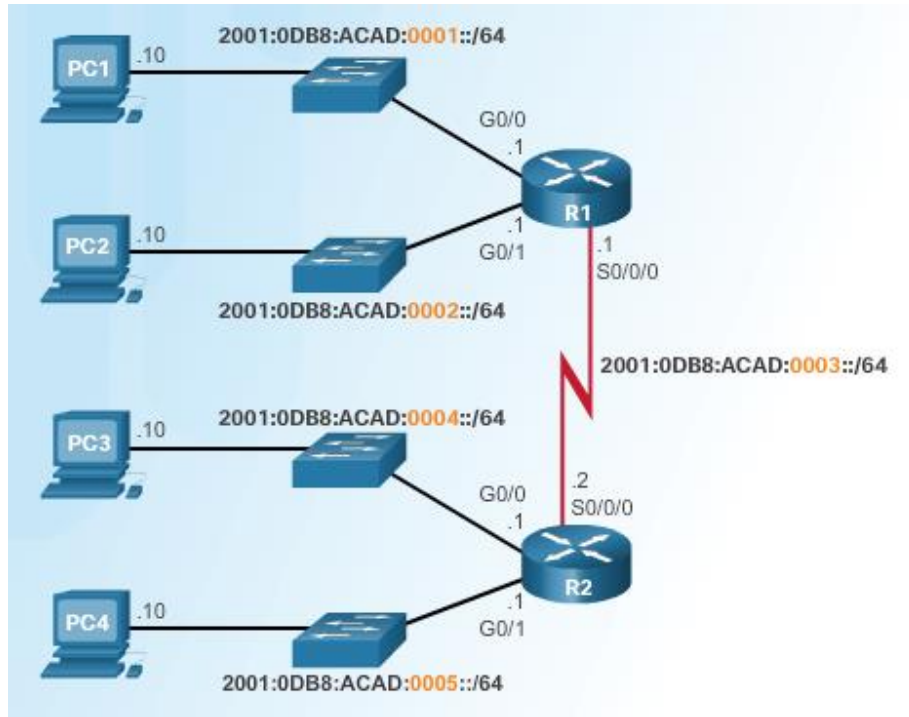
# Subnetting Using the Subnet ID





# Subnetting an IPv6 Network

## IPv6 Subnet Allocation



Address Block: 2001:0DB8:ACAD::/48

5 subnets allocated  
from 65,536  
available subnets

```
2001:0DB8:ACAD:0000::/64
2001:0DB8:ACAD:0001::/64
2001:0DB8:ACAD:0002::/64
2001:0DB8:ACAD:0003::/64
2001:0DB8:ACAD:0004::/64
2001:0DB8:ACAD:0005::/64
2001:0DB8:ACAD:0006::/64
2001:0DB8:ACAD:0007::/64
2001:0DB8:ACAD:0008::/64
2001:0DB8:ACAD:FFFF::/64
```



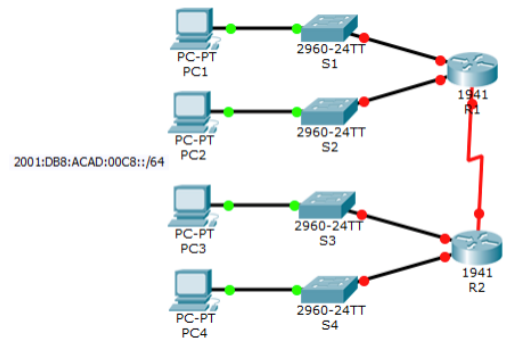
# Subnetting an IPv6 Network

## Packet Tracer – Implementing a Subnetted IPv6 Addressing Scheme



### Packet Tracer - Implementing a Subnetted IPv6 Addressing Scheme

#### Topology



#### Addressing Table

| Device | Interface | IPv6 Address | Link-Local |
|--------|-----------|--------------|------------|
| R1     | G0/0      |              | FE80::1    |
|        | G0/1      |              | FE80::1    |
|        | S0/0/0    |              | FE80::1    |
| R2     | G0/0      |              | FE80::2    |
|        | G0/1      |              | FE80::2    |
|        | S0/0/0    |              | FE80::2    |
| PC1    | NIC       | Auto Config  |            |
| PC2    | NIC       | Auto Config  |            |

# 8.4 Chapter Summary

# Packet Tracer - Skills Integration Challenge

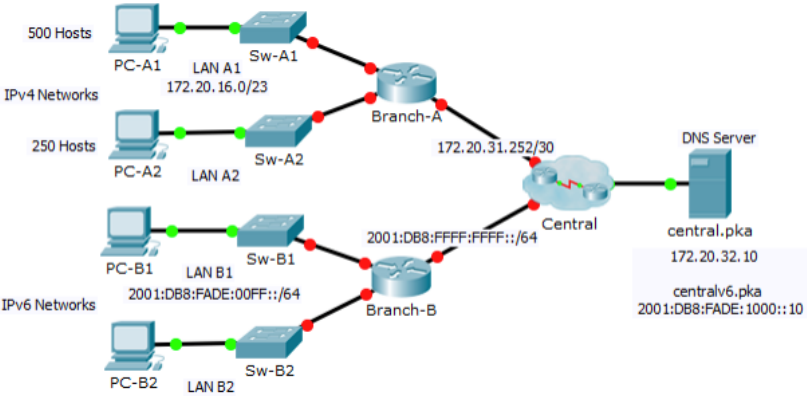


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### Packet Tracer - Skills Integration Challenge

#### Topology



500 Hosts  
PC-A1 LAN A1 Sw-A1  
IPv4 Networks  
250 Hosts  
PC-A2 LAN A2 Sw-A2  
PC-B1 LAN B1 Sw-B1  
IPv6 Networks  
PC-B2 LAN B2 Sw-B2

Branch-A  
Branch-B  
Central  
DNS Server  
central.pk  
172.20.32.10  
centralv6.pk  
2001:DB8:FADE:1000::10

172.20.16.0/23  
2001:DB8:FADE:00FF::/64  
2001:DB8:FFFF:FFFF::/64  
172.20.31.252/30

#### Addressing Table

| Device   | Interface | IPv4 Address        | Subnet Mask     | Default Gateway |
|----------|-----------|---------------------|-----------------|-----------------|
|          |           | IPv6 Address/Prefix |                 |                 |
| Branch-A | G0/0      |                     |                 | N/A             |
|          | G0/1      |                     |                 | N/A             |
|          | G0/2      | 172.20.31.254       | 255.255.255.252 | N/A             |
|          | G0/3      |                     |                 | N/A             |

## Chapter 8: Subnetting IP Networks

- Implement an IPv4 addressing scheme to enable end-to-end connectivity in a small to medium-sized business network.
- Given a set of requirements, implement a VLSM addressing scheme to provide connectivity to end users in a small to medium-sized network.
- Explain design considerations for implementing IPv6 in a business network.

